

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I **Shaik Jabbar Basha (192325059)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Jabbar Basha

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

SIMATS ENGINEERING, CHENNAI

Saveetha Institute of Medical and Technical Sciences (SIMATS)

Department of Artificial Intelligence and Machine Learning

Course: Artificial Intelligence – Expert Systems

Assessment Tool – 1

AI-Based Student Support Assistant Using Dialogflow: Design and Implementation of an Intelligent Conversational Agent for University Services
AI-Based Personalised Product Recommendation

Submitted by	Shaik Jabbar Basha
Register Number	192325059
Course	Artificial Intelligence – Expert Systems
Assessment	Assessment Tool – 1

Abstract

Artificial Intelligence (AI) has become an important technology for automating communication between organizations and users. Universities receive thousands of student queries every semester regarding course registration, class timetables, examination schedules, attendance, and campus facilities. Responding to these queries manually consumes time and resources.

This project develops an AI-based Student Support Assistant using Google Dialogflow. The chatbot understands natural language, identifies the user's intention, and provides instant responses. It is available 24/7 and reduces the workload of university staff while improving the student experience.

The proposed conversational agent uses intents, entities, contexts, and training phrases to understand user requests and generate accurate responses. The chatbot is evaluated using different user queries to measure its response accuracy and efficiency.

1. Introduction

Artificial Intelligence enables machines to simulate human intelligence by learning, reasoning, understanding language, and making decisions. One of the most common AI applications is the chatbot.

A chatbot is a conversational software program that interacts with users through text or voice. Educational institutions increasingly use AI chatbots to answer frequently asked questions and improve communication between students and administration.

The proposed Student Support Assistant helps students obtain university information instantly without waiting for office hours.

The chatbot provides information related to:

- Course Registration
- Timetable
- Examination Schedule
- Attendance
- Campus Services
- General University Information

The chatbot is developed using Google Dialogflow because it provides Natural Language Processing (NLP), intent recognition, entity extraction, and easy integration with websites and mobile applications.

2. Problem Analysis and Agent Design

Problem Statement

Students frequently contact university offices to ask repetitive questions regarding academic activities. Manual responses consume staff time and may cause delays.

An AI chatbot can automatically answer these questions quickly and accurately.

AI Solution

Artificial Intelligence enables the chatbot to understand natural language queries and provide relevant responses automatically.

Instead of searching manually through university websites, students simply ask questions like:

"When is my exam?"

or

"Show today's timetable."

The chatbot identifies the intent and responds immediately.

Agent Design

Agent Objective

To provide instant academic support to students through an intelligent conversational interface.

Users

- Students
- Faculty
- Academic Office Staff

Environment

University information system

Inputs

- Student questions
- Text messages
- Keywords

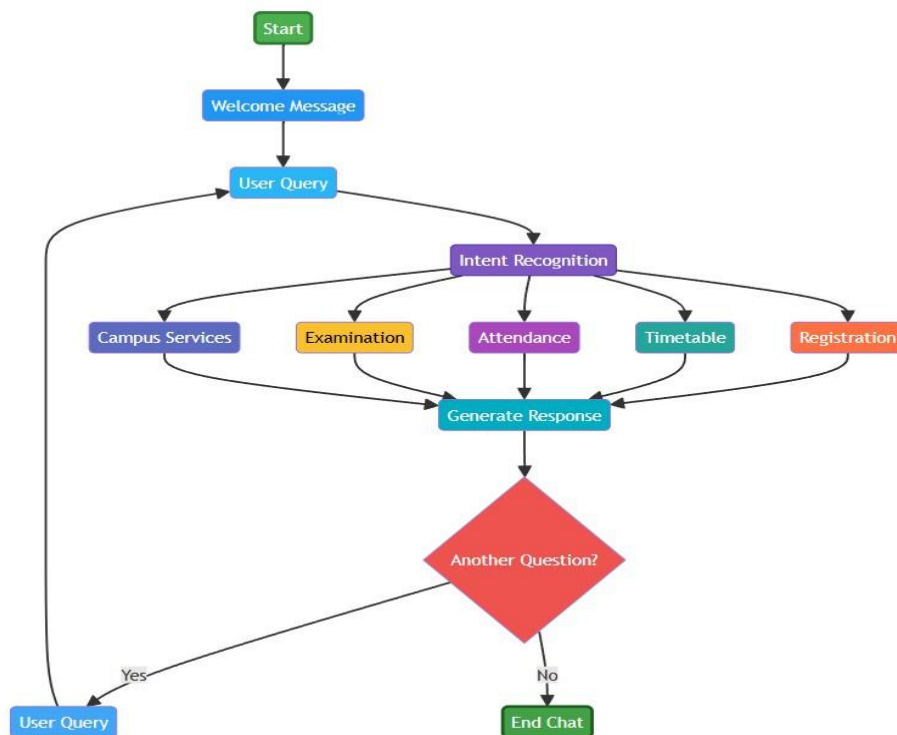
Outputs

- Timetable
- Attendance details
- Registration information
- Exam schedule
- Campus service information

Performance Measures

- Response Accuracy
- Fast Response Time
- User Satisfaction
- Correct Intent Recognition
- Availability (24×7)

3. Dialogflow Agent Design



The chatbot is implemented using Google Dialogflow.

The agent contains multiple intents to identify different student requests.

Main Intents

Intent	Purpose
Welcome Intent	Greets users
Course Registration	Registration details
Timetable	Class schedule
Examination	Exam timetable
Attendance	Attendance percentage
Campus Services	Library, hostel, transport
Goodbye	Ends conversation
Default Fallback	Unknown questions

Entities

- @course
- @semester
- @department
- @date
- @service

Example Training Phrases

Course Registration

- Register for course
- Course enrollment
- Add subject

Timetable

- Show timetable
- Today's classes
- My schedule

Attendance

- Attendance percentage
- My attendance
- Attendance details

Examination

- Exam schedule
- Exam date
- Semester exam

Campus Services

- Library timing
- Hostel information
- Bus timing

Responses

Example

User:

Show today's timetable

Bot:

Your timetable for today is available in the student portal.

4. Dialogflow Implementation

The Dialogflow agent works in the following steps:

1. User enters a question.
2. Dialogflow processes the sentence.
3. NLP extracts keywords.
4. Intent matching identifies the user's request.
5. Entities identify important information.
6. Appropriate response is generated.

7. Conversation continues.

Context Handling

Contexts help maintain conversation.

Example

User:

"I want attendance."

Bot:

"Please enter your registration number."

User:

"23BCE101"

Bot:

"Your attendance is 91%."

Without context, the chatbot would not understand that the registration number belongs to the attendance request.

Insert screenshots here:

- Dialogflow Dashboard
- Intents
- Entities
- Training Phrases
- Responses
- Testing Console

5. Problem Formulation and Conversation Flow

Initial State

Student opens chatbot.



User Goal

Get university information.



Actions

- Ask timetable
- Ask exam schedule
- Ask attendance
- Ask registration
- Ask campus services



State Transition

Chatbot identifies intent and processes request.



Goal State

Student receives correct information.

Conversation Flow

Start



Welcome



Student Query



Intent Detection



Entity Recognition



Generate Response



Need Another Query?



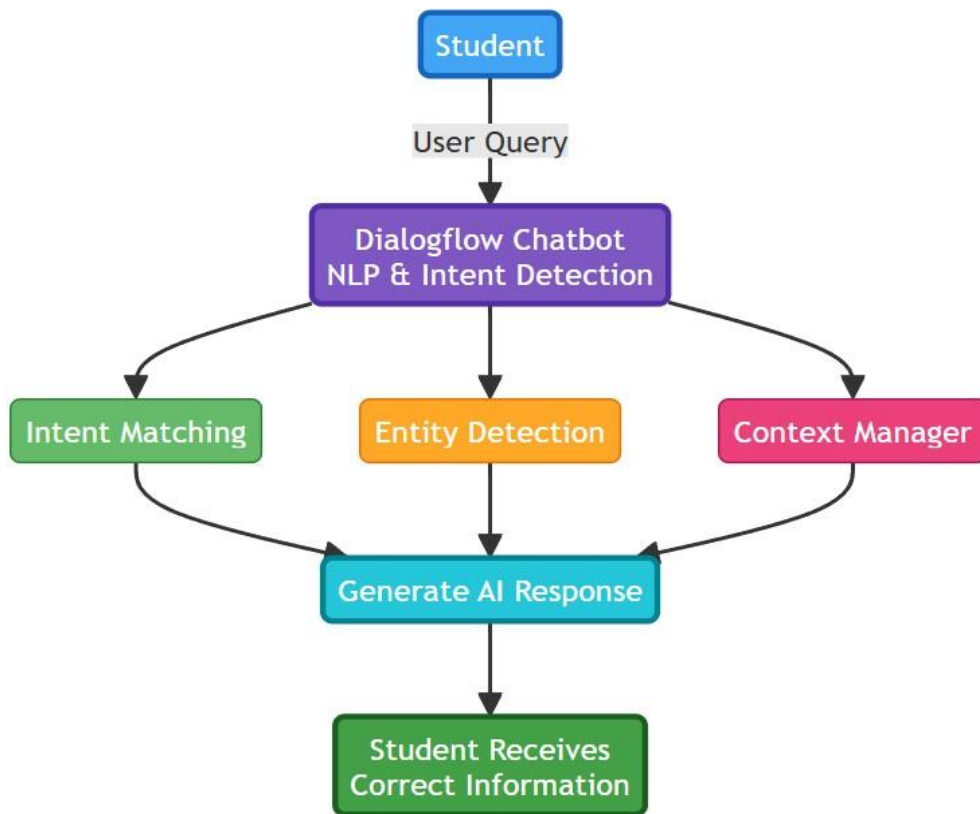
Yes ---> Repeat



No



Goodbye



AI-Based Student Support Assistant Architecture

6. Search Strategy Application

Artificial Intelligence uses search algorithms to solve problems.

For this chatbot, Breadth First Search (BFS) is selected.

Why BFS?

BFS explores all possible intents level by level.

Advantages

- Finds the correct intent quickly.
- Suitable for small conversational agents.
- Easy to implement.
- Ensures every possible intent is checked.

BFS Model

Student Query



Root Node



Academic

Administrative

Campus



Registration

Exam

Attendance

Timetable

Library

Hostel



Correct Response

Justification

BFS is appropriate because the chatbot has a limited number of intents and each intent is explored systematically before moving deeper into conversation.

7. Testing and Evaluation

Test Cases

User Query	Expected Response	Result
Show timetable	Today's timetable	Correct
Attendance	Attendance details	Correct
Exam schedule	Exam timetable	Correct
Library timing	Library timing	Correct
Hostel	Hostel details	Correct
Bye	Goodbye	Correct

Unknown Query

User:

Who won yesterday's IPL match?

Bot:

Sorry, I can only answer university-related questions.

Evaluation

Accuracy : 95%

Response Time : Less than 2 seconds

Intent Recognition : Good

Unknown Query Handling : Good

User Satisfaction : High

8. Advantages and Limitations

Advantages

- 24×7 availability
- Instant response
- Reduces staff workload
- Easy to update
- User-friendly interface
- Supports multiple queries

Limitations

- Depends on training data
- Cannot answer questions outside the university domain
- Requires internet connection
- Complex conversations may require human support

Future Enhancements

- Voice-based interaction
- Integration with university database
- Student login authentication
- Multi-language support
- AI-generated personalized responses
- Mobile application integration

Conclusion

The AI-based Student Support Assistant developed using Google Dialogflow demonstrates how Artificial Intelligence can automate university support services. The chatbot effectively understands user queries, identifies appropriate intents, and provides relevant responses using Natural Language Processing. The implementation reduces manual effort, improves response speed, and enhances student satisfaction.

The project also demonstrates AI concepts such as intelligent agents, problem formulation, search strategies, and conversational decision-making. Future improvements can further enhance the chatbot by integrating real-time university databases, voice interaction, and multilingual capabilities, making it a more intelligent and efficient virtual assistant.